

Spring Core & Maven Exercises

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Exercise 1: Configuring a Basic Spring Application

pom.xml

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.30</version>
  </dependency>
</dependencies>
```

BookRepository.java

```
package com.library.repository;
public class BookRepository {
    public void display() {
        System.out.println("Book Repository Created");
    }
}
```

BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository){
        this.bookRepository=bookRepository;
    }
    public void showService(){
        System.out.println("Book Service Created");
        bookRepository.display();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">
<bean id="bookRepository"
class="com.library.repository.BookRepository"/>
<bean id="bookService"
class="com.library.service.BookService">
<property name="bookRepository" ref="bookRepository"/>
</bean>
</beans>
```

MainApp.java

```
package com.library;
import org.springframework.context.*;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class MainApp{
    public static void main(String[] args){
        ApplicationContext context=new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service=(BookService)context.getBean("bookService");
        service.showService();
    }
}
```

Output

```
Book Service Created
Book Repository Created
```

Exercise 2: Implementing Dependency Injection

BookRepository.java

```
package com.library.repository;
public class BookRepository{
    public void display(){
        System.out.println("Book Repository Connected");
    }
}
```

BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService{
    private BookRepository bookRepository;
    public void setBookRepository(BookRepository bookRepository){
        this.bookRepository=bookRepository;
    }
    public void showService(){
        System.out.println("Book Service Running");
        bookRepository.display();
    }
}
```

applicationContext.xml

```
<bean id="bookRepository" class="com.library.repository.BookRepository"/>
<bean id="bookService" class="com.library.service.BookService">
    <property name="bookRepository" ref="bookRepository"/>
</bean>
```

LibraryManagementApplication.java

```
ApplicationContext context=
new ClassPathXmlApplicationContext("applicationContext.xml");
BookService service=(BookService)context.getBean("bookService");
service.showService();
```

Output

```
Book Service Running
Book Repository Connected
```

Exercise 3: Creating and Configuring a Maven Project

pom.xml

```
<project>
  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0</version>
  <dependencies>
    spring-context 5.3.30
    spring-aop 5.3.30
    spring-webmvc 5.3.30
  </dependencies>
  <build>
    maven-compiler-plugin 3.8.1
    source=1.8 target=1.8
  </build>
</project>
```

Project Structure

```
LibraryManagement
|-- src/main/java
|-- src/main/resources
`-- pom.xml
```

Maven Commands

```
mvn clean
mvn compile
mvn package
```

Output

```
BUILD SUCCESS
```